Nested For Loops

Pattern Programming Exercises · C#

Instructions

- Use nested **for** loops to solve all the questions.
- Take the number of rows (n) as input wherever required.
- Do not use built-in pattern-printing methods.

1. Right-Angled Star Triangle

Write a C# program to print a right-angled triangle using stars (*).

Example (n = 5):

```
*
**
***
****
*****
```

2. Inverted Right-Angled Triangle

Write a program to print an inverted right-angled triangle.

Example (n = 5):

```
*****
****
***
**
*
```

3. Right-Aligned Triangle

Write a program to print a right-aligned triangle using stars.

Example (n = 5):

```
    *
   **
  ***
 ****
*****
```

4. Pyramid Pattern

Write a program to print a centered pyramid using stars.

Example (n = 5):

```
    *
   ***
  *****
 *******
*********
```

5. Inverted Pyramid

Write a program to print an inverted centered pyramid.

Example (n = 5):

```
*********
 *******
  *****
   ***
    *
```

6. Diamond Pattern

Write a program to print a diamond shape using stars.

Example (n = 5):

```
    *
   ***
  *****
 *******
*********
 *******
  *****
   ***
    *
```

7. Hollow Square Pattern

Write a program to print a hollow square using stars. Only the boundary should contain stars.

Example (n = 5):

```
*****
*   *
*   *
*   *
*****
```

8. Number Triangle

Write a program to print numbers in a right-angled triangle. Each row should start printing from 1.

Example (n = 5):

```
1
12
123
1234
12345
```

9. Floyd's Triangle

Write a program to print Floyd's Triangle. Numbers should continue sequentially from one row to the next.

Example (n = 5):

```
1
2 3
4 5 6
7 8 9 10
11 12 13 14 15
```

10. Butterfly Pattern

Write a program to print a butterfly pattern using stars.

Example (n = 5):

```
*         *
**       **
***     ***
****   ****
**********
****   ****
***     ***
**       **
*         *
```